Road Map

FACE_VERTEX: $FV(f,v) = \begin{cases} 1 & \text{if } v \notin f \\ 0 & \end{cases}$

Support of Slack matrix encodes FV.

$\text{Supp}(S) = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ & & 1 & 0 & 1 \end{bmatrix}$

only special facets: F_A

facets

vertices: $y^B = x^B x^{B^T}$, $B \subseteq [n]$

$FV(A,B)$ A

$A \subseteq [n] \to F_A$

$B \subseteq [n] \to y^B$

Make easier still: Because of properties of F_A: $y_B \in F_A \iff |A \cap B| = 1$

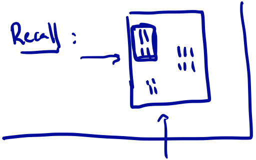
$\iff FV(A,B) = 1$ iff $|A \cap B| \neq 1$

Promise: $|A \cap B| \in \{0,1\}$

Valid Sets and Covering D(n)

$$\mathcal{D}(n) = \{ (A,B) \in 2^{[n]} \times 2^{[n]} : A \cap B = \phi \}$$

Recall:



Def'n: $R \subseteq \mathcal{D}(n)$ is called **valid** if $\forall (A,B), (A',B') \in R$

$|A \cap B'| \neq 1$, $|A' \cap B| \neq 1$.

Suppose we have a covering of $\mathcal{D}(n) \subseteq R_1 \cup R_2 \cup \dots \cup R_r$, R_i is valid.

1. Alice receives A, Bob receives B

   Prover sees (A,B)

2. If $F_V(A,B) = 1$ then $A \cap B = \phi \Rightarrow (A,B) \in \mathcal{D}(n) \Rightarrow (A,B) \in R_i$.

   Prover ~~k~~ sends index i to Alice & Bob

3. Alice checks if $\exists \tilde{B}$ s.t. $(A, \tilde{B}) \in R_i$ if so Alice accepts

   Bob " $\exists \tilde{A}$ $(\tilde{A}, B) \in R_i$ Bob accepts.

R_i valid : $(A,\tilde{B}) \in R_i$, $(\tilde{A}, B) \in R_i \Rightarrow |A \cap B| = 0$ ✓

Extended Formulations Give Coverings

Lemma: If Q is an extended formulation for P_{corr} with r facets (i.e., $\pi(Q) = P_{corr}$), then $\exists$ covering of $\mathcal{D}(n)$ of size r. (with valid sets)

Step 1: prove this lemma

Step 2: show any covering requires at least 1.5^n sets (valid).

Proof — 1/2

Q extended formulation. $\pi(Q) = P_{corr}$

Facets: $G_1, \ldots, G_r$. We use these to define valid sets

R_i, and then we show these form a covering.

For any facet G of Q, define the set:

$$R_G \doteq \{(A,B) \in \mathcal{D}(n) : \text{(i) } \underline{\pi^{-1}(F_A) \cap Q} \subseteq G \quad \text{(ii) } \pi^{-1}(y^B) \cap Q \nsubseteq G\}$$

Need to prove: Claim 1: R_G valid, Claim 2: $\mathcal{D}(n) \subseteq \bigcup_{i=1}^{r} R_{G_i}$

<u>Proof of Claim 1</u>: Need: $(A,B), (A',B') \in R_G \Rightarrow |A \cap B'| \neq 1$

For if $|A \cap B'| = 1 \Leftrightarrow y^{B'} \in F_A \Rightarrow \underline{\pi^{-1}(y^{B'}) \cap Q} \subseteq \pi^{-1}(F_A) \cap Q \subseteq G$ violating (ii).

Proof — 2/2

proof of claim 2: $\cup R_{G_i}$ is a covering of $\mathcal{D}(n)$.

Take $(\hat{A}, \hat{B}) \in \mathcal{D}(n)$ so $\hat{A} \cap \hat{B} = \phi$.

$F_{\hat{A}}$ face of $P_{corr.}$ $\quad \pi^{-1}(F_{\hat{A}}) \cap Q$ is a face of Q.

Since any face of a polytope is the intersection of facets that contain it: $\underline{\pi^{-1}(F_{\hat{A}}) \cap Q} = G_{i_1} \cap G_{i_2} \cap \cdots \cap G_{i_k}$.

$$\hat{A} \cap \hat{B} = \phi \Rightarrow y^{\hat{B}} \notin F_{\hat{A}}$$

$$\Rightarrow \underline{\pi^{-1}(y^{\hat{B}}) \cap Q} \notin \underline{\pi^{-1}(F_{\hat{A}}) \cap Q} = G_{i_1} \cap G_{i_2} \cap \cdots \cap G_{i_k}$$

Hence, $\exists$ some i_ℓ for which: $\pi^{-1}(y^{\hat{B}}) \cap Q \notin G_{i_\ell}$ — condition (ii) —

since $\pi^{-1}(F_{\hat{A}}) \cap Q = \bigcap G_{i_j} \Rightarrow \pi^{-1}(F_{\hat{A}}) \cap Q \subseteq G_{i_\ell}$

— condition (i) —

Hence: $(\hat{A}, \hat{B}) \in R_{G_{i_\ell}}$ $\quad$ hence $\mathcal{D}(n) \subseteq R_{G_1} \cup \cdots \cup R_{G_r}$.